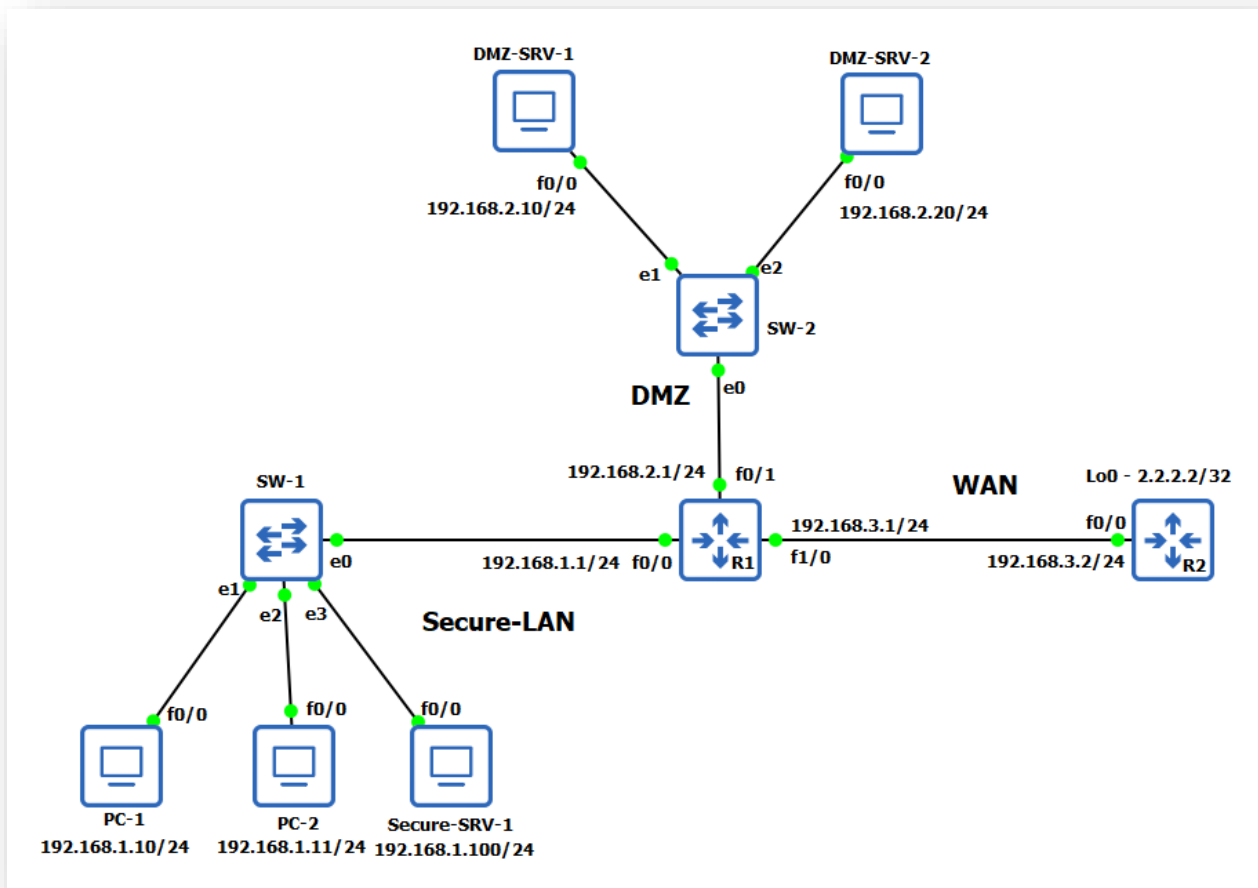


Extended ACL – Number & Named



Task

1. Configure routers R1 & R2 with IP address as shown in topology and configure enable password as **ccna**. (**Refer Lab-16 for Task-1 to Task-5**)
2. Configure routers as Host (PC / Servers)
3. On Secure-SRV-1, enable SSH and HTTPs service on port 1025.
4. On DMZ-SRV-1 & DMZ-SRV-1, enable SSH and HTTP service.
5. Configure default routing to provide connectivity between them.
6. Configure extended number ACL to block ping to Secure-SRV-1 from DMZ and DMZ-SRV-1 can access Secure-SRV-1 server via HTTPs on port 1025.
7. Configure extended named ACL to block ping to Secure-SRV-1 from DMZ but Secure-SRV-1 can ping DMZ servers and DMZ-SRV-1 should can access Secure-SRV-1 server via HTTPs on port 1025.

Task-6: Configure extended number ACL to block ping to Secure-SRV-1 from DMZ and DMZ-SRV-1 can access Secure-SRV-1 server via HTTPs on port 1025.

Source: DMZ network

Destination: Secure-LAN

Implementation: R1 (Close to Source)

Traffic inspection: Inbound traffic

```

R1#config t
R1(config)# access-list 100 permit tcp host 192.168.2.10 host 192.168.1.100 eq 1025 log
R1(config)#access-list 100 deny tcp any host 192.168.1.100 eq 1025 log
R1(config)# access-list 100 deny icmp 192.168.2.0 0.0.0.255 host 192.168.1.100 log
R1(config)#access-list 100 permit ip any any
R1(config)#
R1(config)#int fa0/1
R1(config-if)#ip access-group 100 in
R1(config-if)#exit
R1(config)#exit
R1#

```

✓ Verification & Testing:

R1#sh ip access-lists

Extended IP access list 100

```

10 permit tcp host 192.168.2.10 host 192.168.1.100 eq 1025 log
20 deny tcp any host 192.168.1.100 eq 1025 log
30 deny icmp 192.168.2.0 0.0.0.255 host 192.168.1.100 log
40 permit ip any any

```

R1#

Test connectivity and HTTPs services to Secure-LAN PCs and servers from DMZ-SRV-1

DMZ-SRV-1#ping 192.168.1.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/40/40 ms

DMZ-SRV-1#ping 192.168.1.11

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.11, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 36/40/44 ms

DMZ-SRV-1#ping 192.168.1.100

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.100, timeout is 2 seconds:

UUUUU

Success rate is 0 percent (0/5)

DMZ-SRV-1#telnet 192.168.1.100 1025

Trying 192.168.1.100, 1025 ... Open

DMZ-SRV-1#ping 2.2.2.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/42/48 ms

R1#

*Nov 5 22:04:22.047: %SEC-6-IPACCESSLOGDP: list 100 denied icmp 192.168.2.10 -> 192.168.1.100 (8/0), 5 packets

R1#

*Nov 5 22:04:23.587: %SEC-6-IPACCESSLOGP: list 100 permitted tcp 192.168.2.10(33218) -> 192.168.1.100(1025), 1 packet

R1#

Test connectivity and HTTPs services to Secure-LAN PCs and servers from DMZ-SRV-2

DMZ-SRV-2#ping 192.168.1.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 8/38/64 ms

DMZ-SRV-2#ping 192.168.1.11

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.11, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 16/40/64 ms

DMZ-SRV-2#ping 192.168.1.100

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.100, timeout is 2 seconds:

UUUUU

Success rate is 0 percent (0/5)

DMZ-SRV-2#telnet 192.168.1.100 1025

Trying 192.168.1.100, 1025 ...

% Destination unreachable; gateway or host down

DMZ-SRV-2#ping 2.2.2.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 36/39/40 ms

DMZ-SRV-2#

R1#

*Nov 5 22:06:37.023: %SEC-6-IPACCESSLOGP: list 100 denied tcp 192.168.2.20(13524) -> 192.168.1.100(1025), 1 packet

R1#

*Nov 5 22:07:22.051: %SEC-6-IPACCESSLOGDP: list 100 denied icmp 192.168.2.20 -> 192.168.1.100 (8/0), 10 packets

R1#

Test connectivity from Secure-LAN PCs and servers to DMZ network

PC-1#ping 192.168.2.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/39/64 ms

PC-1#ping 192.168.2.20

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.20, timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/40/40 ms

PC-2#ping 192.168.2.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 32/38/40 ms

PC-2#ping 192.168.2.20

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.20, timeout is 2 seconds:

!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 8/38/68 ms

Secure-SRV-1#ping 192.168.2.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

.....

Success rate is 0 percent (0/5)

Secure-SRV-1#ping 192.168.2.20

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.20, timeout is 2 seconds:

.....

Success rate is 0 percent (0/5)

To Remove ACL

R1#config t

R1(config)#no access-list 100

R1(config)#int fa0/1

R1(config-if)#no ip access-group 100 in

R1(config-if)#exit

R1(config)#exit

R1#

Task-7: Configure extended named ACL to block ping to Secure-SRV-1 from DMZ but Secure-SRV-1 can ping DMZ servers and DMZ-SRV-1 should can access Secure-SRV-1 server via HTTPs on port 1025.

Source: DMZ network

Destination: Secure-LAN network

Implementation: R1 (Close to Source)

Traffic inspection: Outbound traffic

```
R1#config t
R1(config)#ip access-list extended YourACL
R1(config-ext-nacl)#deny icmp 192.168.2.0 0.0.0.255 host 192.168.1.100 echo log
R1(config-ext-nacl)#deny tcp host 192.168.2.10 host 192.168.1.100 eq 1025 log
R1(config-ext-nacl)#permit tcp any host 192.168.1.100 eq 1025 log
R1(config-ext-nacl)#permit ip any any
R1(config-ext-nacl)#exit
R1(config)#int fa0/0
R1(config-if)#ip access-group YourACL out
R1(config-if)#exit
R1(config)#exit
R1#
```

✓ Verification & Testing:

R1#sh run | sec access-list

```
ip access-list extended YourACL
deny icmp 192.168.2.0 0.0.0.255 host 192.168.1.100 echo log
deny tcp host 192.168.2.10 host 192.168.1.100 eq 1025 log
permit tcp any host 192.168.1.100 eq 1025 log
permit ip any any
R1#
```

R1#sh access-lists

```
Extended IP access list YourACL
 10 deny icmp 192.168.2.0 0.0.0.255 host 192.168.1.100 echo log
 20 deny tcp host 192.168.2.10 host 192.168.1.100 eq 1025 log
 30 permit tcp any host 192.168.1.100 eq 1025 log
 40 permit ip any any
R1#
```

Test connectivity and HTTPs services to Secure-LAN PCs and servers from DMZ-SRV-1

DMZ-SRV-1#ping 192.168.1.10

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 40/71/148 ms
```

DMZ-SRV-1#ping 192.168.1.11

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.11, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 40/46/64 ms
```

DMZ-SRV-1#ping 192.168.1.100

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.100, timeout is 2 seconds:
.UUUU
Success rate is 0 percent (0/5)
```

DMZ-SRV-1#telnet 192.168.1.100 1025

Trying 192.168.1.100, 1025 ...

% Destination unreachable; gateway or host down

R1#

*Nov 5 09:34:30.919: %SEC-6-IPACCESSLOGDP: list YourACL denied icmp 192.168.2.10 -> 192.168.1.100 (8/0), 1 packet

R1#

*Nov 5 09:34:44.527: %SEC-6-IPACCESSLOGDP: list YourACL denied tcp 192.168.2.10(26888) -> 192.168.1.100(1025), 1 packet

R1#

Test connectivity and HTTPs services to Secure-LAN PCs and servers from DMZ-SRV-2

DMZ-SRV-2#ping 192.168.1.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 112/333/1092 ms

DMZ-SRV-2#ping 192.168.1.11

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.11, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 36/40/44 ms

DMZ-SRV-2#ping 192.168.1.100

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.100, timeout is 2 seconds:

UUUUU

Success rate is 0 percent (0/5)

DMZ-SRV-2#telnet 192.168.1.100 1025

Trying 192.168.1.100, 1025 ... Open

DMZ-SRV-2#

Test connectivity from Secure-LAN PCs and servers to DMZ network

PC-1#ping 192.168.2.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 36/40/44 ms

PC-1#ping 192.168.2.20

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.20, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 28/40/52 ms

PC-2#ping 192.168.2.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/40/44 ms

PC-2#ping 192.168.2.20

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.20, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 36/40/44 ms

Secure-SRV-1#ping 192.168.2.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 16/39/60 ms

Secure-SRV-1#ping 192.168.2.20

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.20, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/40/40 ms

Secure-SRV-1#

R1#

*Nov 5 09:36:01.735: %SEC-6-IPACCESSLOGDP: list YourACL denied icmp 192.168.2.20 -> 192.168.1.100 (8/0), 1 packet

R1#

*Nov 5 09:36:14.483: %SEC-6-IPACCESSLOGP: list YourACL permitted tcp 192.168.2.20(59191) -> 192.168.1.100(1025), 1 packet

R1#

To Remove ACL

R1#config t

R1(config)#no ip access-list extended YourACL

R1(config)#int fa0/0

R1(config-if)#no ip access-group YourACL out

R1(config-if)#exit

R1(config)#exit

R1#

Important Commands:

```
sh access-lists  
sh ip access-lists  
sh run | sec access-list
```

Ratan